Topic-Aware LLM Framework for NEET/JEE Chemistry Question Generation Using Knowledge Graph and Multi-Agent System

1 Abstract

A knowledge graph and multi-agent large language model (LLM) framework for topic-aware chemistry question generation is proposed to address the limitations of existing LLM-based systems, which may generate irrelevant, repetitive, or factually incorrect questions. The proposed framework integrates a knowledge graph, a five-agent system, and an LLM to generate high-quality Chemistry multiple-choice questions (MCQs), incorporating chemical structure image generation and similarity checking. Text-based questions are directly processed, whereas image-based questions are denoised and enhanced using a U-Net before question and concept extraction. Multiple agents perform topic selection, knowledge graph retrieval, prompt construction, question generation, Swin Transformer-based image similarity detection, and validation. The framework generates syllabus-aware questions by combining structured knowledge graph information with LLM-based generation. The eQOURSE JEE Main chemistry dataset is used to construct the knowledge graph and provide structured information related to questions and concepts. The proposed framework can support students, teachers, coaching centres, and educational platforms by automating chemistry question generation and reducing the manual effort involved in question preparation.

2 Introduction

The rapid advancement in AI, especially LLMs, has given rise to the possibility of using AI to automate the creation of educational materials. The domain of education may benefit from automated question generation since this technology would help minimize the time spent on creating questions for practice and examination purposes, offering students a wider variety of questions. Chemistry is one of the domains that could greatly benefit from the development of automated question generation since there are many interrelated concepts in Chemistry that include physical, organic, and inorganic chemistry, as well as chemical equations, mathematical formulas, structures, symbols, and diagrams. Nevertheless, creating quality Chemistry questions using automated generation is harder than just producing syntactically correct sentences. An automatically generated question should not only be relevant to the chosen concept but also chemically meaningful, properly structured, and appropriate for a specific difficulty level.

The current project deals with automatic Chemistry questions generation through the use of Knowledge Graphs (KGs), Multi-Agent Systems (MAS), Large Language Models (LLMs), and U-Net. The current body of Chemistry questions serves as the main source of domain knowledge. Such questions are processed and represented in the form of a structured Knowledge Graph consisting of information on chapters, topics, subtopics, concepts, questions, options, answers, the level of difficulty and their connections. With the representation of available Chemistry knowledge in structured format, the system is able to find out key concepts of a chosen topic, as well as related questions and knowledge prior to generation stage. Thus, the system provides a source of controlled knowledge for the upcoming question generation process. Rather than relying solely on the knowledge that is embedded in an LLM, the current framework enables generation process to be driven by domain-specific knowledge extracted from the Chemistry Knowledge Graph.

Even though LLMs have shown proficiency in understanding and generating natural language, the direct usage of an LLM in generating educational questions will lead to various constraints. The generated

question might be off-topic, repetitive, poorly structured, or may even have factual inaccuracies. Furthermore, there could be a question generated by the LLM which appears grammatically correct but fails to correctly represent a Chemistry concept. All the aforementioned limitations make the problem particularly difficult when used in examination oriented application like JEE/NEET Main where generated questions should meet certain criteria of being on topic, appropriate difficulty level, options, correctness, and relevancy to the concept. Moreover, in some cases, Chemistry questions are not purely in the form of text. They could include images that consist of chemical structures, equations, mathematical expressions, diagrams, symbols, etc. Such images may have degradation or noise in them which can degrade the quality of the information that we obtain from them.

The proposed framework further uses a Multi-Agent System to divide the question-generation workflow into specialized stages rather than assigning the entire task to a single model. The agents are designed to perform specific functions such as topic selection, Knowledge Graph retrieval, concept extraction, difficulty planning, prompt building, question generation, similarity checking, and final validation. The retrieval agent obtains relevant Chemistry information from the Knowledge Graph based on the selected topic, while the concept extraction agent identifies the important concepts associated with the retrieved information. The difficulty agent helps determine the required difficulty level, and the prompt-building agent combines the retrieved concepts and constraints into suitable contextual information for the LLM. A similarity-checking component can compare the newly generated question with existing questions to identify repetitive or highly similar content. Finally, a validation component evaluates the generated question against requirements such as topic relevance, structural correctness, answer consistency, and overall quality. Thus, the proposed system aims to provide a topic-aware, structured, and controlled Chemistry question generation framework in which Knowledge Graph-based retrieval, image preprocessing, specialized agents, and LLM-based generation work together to reduce the limitations of unrestricted question generation.

3 Literature Survey

Automatic Question Generation (AQG) reduces the manual effort required to prepare practice and examination questions. Recent research has progressed from rule-based approaches to neural networks, transformer-based models, retrieval-augmented generation, Knowledge Graphs, and multi-agent systems. The following works are grouped based on the main methodologies used.

Automatic Question Generation:

Das et al. [1] reviewed neural network and transformer-based approaches for automatic question generation and answer assessment using textual and pictorial resources. Mulla and Gharpure [2] surveyed rule-based, neural, and transformer-based methods using models such as BERT, GPT, and T5. These studies discussed commonly used datasets, evaluation metrics, and challenges in generating high-quality questions.

Retrieval-Augmented Generation (RAG):

Lewis et al. [3] introduced Retrieval-Augmented Generation (RAG), which combines a neural retriever with a sequence-to-sequence generator to retrieve relevant external knowledge before generation. The approach improved knowledge-intensive generation by grounding the generated output in retrieved information. This methodology provides a foundation for using external domain knowledge instead of relying only on the knowledge stored within an LLM.

Knowledge Graph-Based Retrieval and Generation:

Edge et al. [4] proposed GraphRAG, which uses an entity-based Knowledge Graph and community summaries to retrieve and organize relevant information before LLM generation, improving the comprehensiveness and diversity of generated responses. Zhu et al. [5] proposed KG^{2}RAG, which combines semantic retrieval with Knowledge Graph-based expansion before generation, improving the retrieval and generation process. These approaches demonstrate how structured knowledge can be combined with retrieval to provide more relevant and context-aware generation.

Multi-Agent and Neuro-Symbolic Approaches:

Wang et al. [7] surveyed LLM-based autonomous agents that use planning, memory, and tool usage to support complex tasks. Hitzler et al. [6] reviewed neuro-symbolic approaches that combine neural

learning with symbolic knowledge representation and reasoning, providing a foundation for integrating learned models with structured knowledge. These approaches support the use of specialized components and structured knowledge in complex AI generation tasks.

Distractor Generation:

Qiu et al. [12] proposed EDGE, an attention-based approach for generating plausible distractors for multiple-choice questions. The method uses question and passage reforming followed by attention-based generation to produce meaningful incorrect options and achieved strong performance compared with existing distractor-generation methods.

4 Dataset Description

The current implementation uses the `eQOURSE/jee-main-questions` chemistry split. After preprocessing, each record contains the following schema attributes:

`Question_ID`	`Question`	`Option_A`	`Option_B`	`Option_C`	`Option_D`	`Answer`	`Difficulty`	`Topic`	`Chapter`

The notebook implementation demonstrates the end-to-end preprocessing and graph-building workflow using Python libraries including `datasets`, `pandas`, `networkx`, and `matplotlib`.

It produces: (1) `cleaned_chemistry_dataset.csv`, (2) a full or subset knowledge graph with typed nodes and relations, and (3) visual graph inspection for sample questions. This implementation confirms that the top-down graph schema is feasible and that graph traversal can expose the chapter–topic–question–option structure for agent retrieval.

5 Methodology

A. System Overview

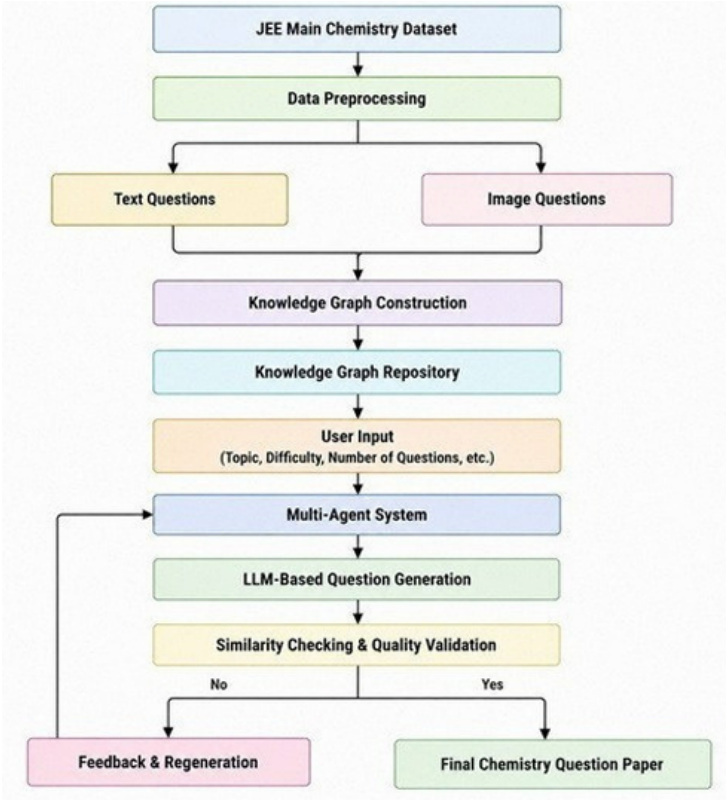


Figure 1: *End-to-end system overview, from the JEE Main Chemistry dataset through preprocessing, Knowledge Graph construction, the Multi-Agent System, and LLM-based generation to the final validated Chemistry question paper.*

The proposed framework follows a four-stage pipeline: (1) dataset acquisition and preprocessing, (2) top-down knowledge-graph construction, (3) multi-agent orchestration for topic selection, retrieval, drafting, and verification, and (4) LLM-based generation of NEET/JEE-style MCQs. Structured knowledge flows from the graph to the agents and then to the LLM, ensuring that generated questions remain anchored to syllabus entities rather than free-form model hallucination.

B. Dataset Acquisition and Preprocessing

The project uses the public Hugging Face dataset `eQOURSE/jee-main-questions` with the Chemistry split. Each record contains question text, four options, the correct option, difficulty, topic, and subtopic metadata. The preprocessing pipeline implemented in the project notebook performs the following steps:

- Load the Chemistry split using the Hugging Face datasets library.
- Convert the dataset to a tabular format and inspect missing values.
- Remove duplicate questions based on question text.
- Exclude image-based questions (`has_image = False`) to keep the initial pipeline text-only.

This preprocessing stage ensures that downstream graph construction operates on deduplicated, text-only, schema-consistent records aligned with the NEET/JEE Chemistry exam format.

C. U-Net Architecture for Chemistry Image Denoising

U-Net is a convolutional neural network used to denoise and enhance Chemistry images. The architecture consists of three main components: an encoder, a bottleneck, and a decoder.

Encoder: Three DoubleConv blocks are used, with channel transitions of $1 \rightarrow 64$, $64 \rightarrow 128$, and $128 \rightarrow 256$. Each block contains two 3×3 convolutional layers with ReLU activation. MaxPooling reduces the spatial dimensions between encoder blocks.

Bottleneck: A DoubleConv block with 512 channels extracts deeper image features.

Decoder: Transposed convolution layers perform upsampling, followed by DoubleConv blocks. Skip connections concatenate encoder feature maps with decoder feature maps to preserve spatial information and fine details.

Residual Connection: A 1×1 convolution maps 64 feature channels to one output channel to predict a correction. This correction is added to the input image, and the output is clipped to the range $[0, 1]$.

Training and Loss Function: The model is trained using the Adam optimizer and a Chemistry-aware loss function combining weighted mean squared error (MSE) and an edge-preservation loss based on L1 loss. The weighted MSE assigns greater importance to darker pixels, while the edge loss encourages the preservation of image boundaries.

The final output is a denoised and enhanced Chemistry image for subsequent question and concept extraction.

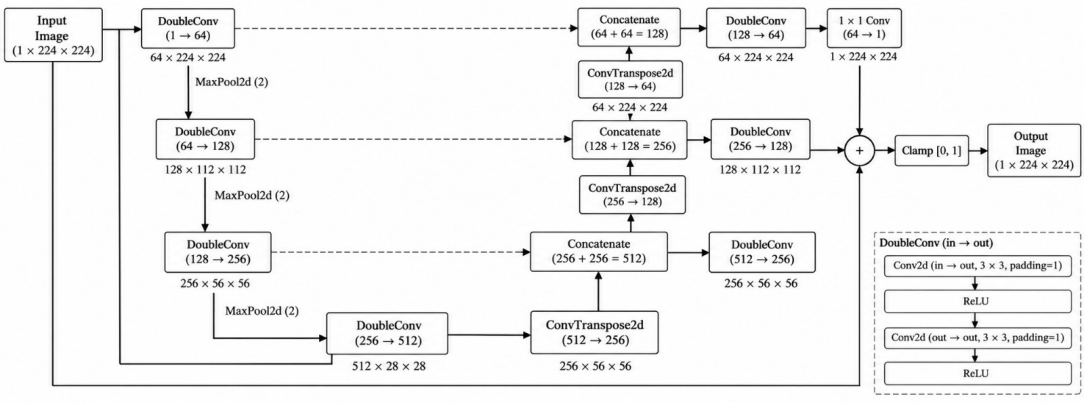


Figure 2: *U-Net Architecture for Chemistry Image Denoising.*

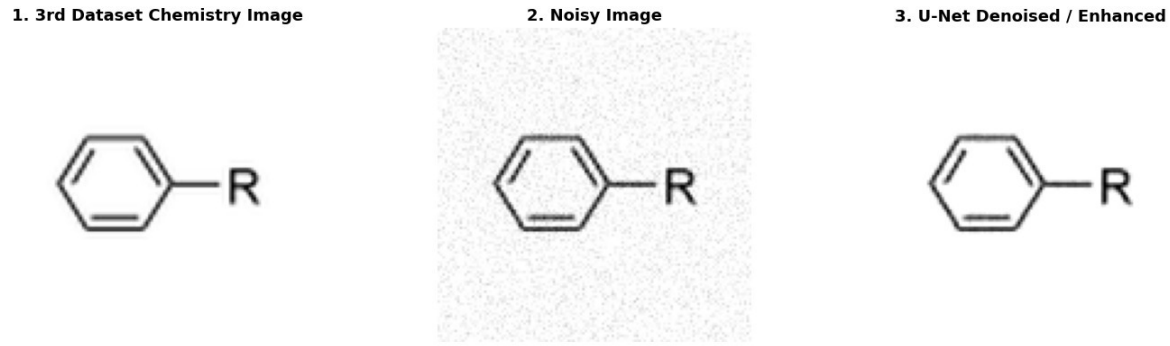


Figure 3: *Chemistry Image Before and After U-Net-Based Denoising.*

D. Top-Down Knowledge Graph Construction

The knowledge graph is constructed using a top-down methodology rather than bottom-up entity extraction. This design choice is deliberate and well matched to the dataset for five reasons.

- The NEET/JEE Chemistry dataset is already organised into structured fields: Chapter, Topic, Question, Options, Answer, and Difficulty.
- Top-down construction yields a consistent and predictable graph schema, which is essential for downstream question generation and retrieval.
- Every question follows the same schema, so predefined node types and relationship labels are sufficient.
- The approach is simpler, faster, and less error-prone than NLP or LLM-based relation extraction on already-structured data.
- It produces uniform relationships throughout the graph, improving retrieval quality and LLM prompting reliability.

By contrast, a bottom-up approach would require entity extraction, relation extraction, duplicate resolution, and validation even though the required information already exists in separate columns. That would increase computational cost and extraction error without providing additional benefit for this project.

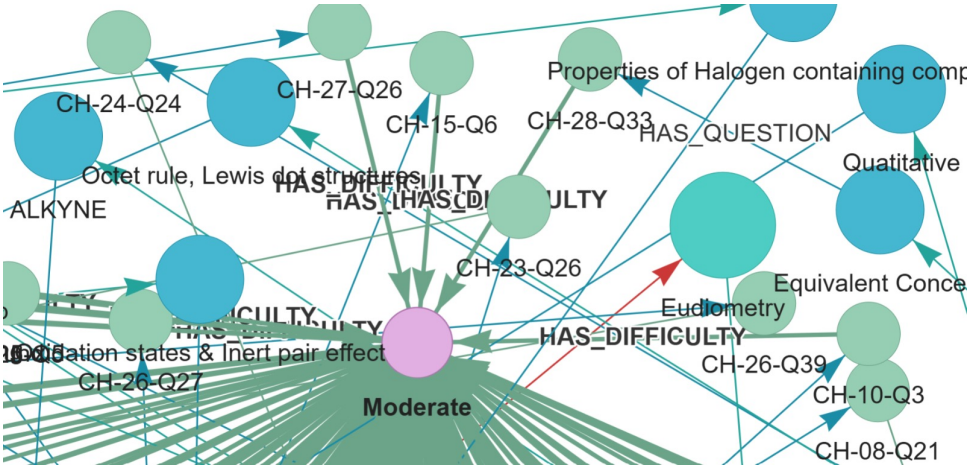


Figure 4: *Partial View of the Chemistry Knowledge Graph..*

E. Graph Schema and Relationship Types

The knowledge graph is implemented as a directed graph using NetworkX. The final schema contains the following node types:

- Chapter – syllabus chapter or subtopic grouping
- Topic – concept-level topic within a chapter
- Question – exam question, referenced by Question ID with full text stored as a node attribute
- Option – one of four MCQ options (A, B, C, D)
- Difficulty – difficulty label associated with a question

The following relationship types connect these nodes:

- Chapter → `HAS_TOPIC` → Topic
- Topic → `HAS_QUESTION` → Question
- Question → `HAS_OPTION` → Option
- Question → `HAS_CORRECT_OPTION` → Option
- Question → `HAS_DIFFICULTY` → Difficulty

In the initial prototype, question nodes use identifiers such as Q{Question_ID} while storing the full question text as an attribute. This prevents node-label clutter in visualisation while preserving retrieval access to the complete question content. Option nodes are prefixed by their letter label (for example, 'A: ...') and the correct option is linked through a dedicated HAS_CORRECT_OPTION edge.

F. Multi-Agent Workflow

The multi-agent system decomposes Chemistry question generation into five specialised agents that work sequentially to generate topic-aware multiple-choice questions (MCQs). Each agent performs a specific task, from analysing user requirements to checking the similarity of generated questions.

1) Requirement Analysis Agent: The Requirement Analysis Agent processes the user's request to identify the required Chemistry topic, subtopic, difficulty level, question type, and number of questions. It converts the request into structured requirements that guide the subsequent agents.

2) Knowledge Graph Retrieval Agent: The Knowledge Graph Retrieval Agent retrieves relevant Chemistry questions, concepts, options, answers, and associated information from the Chemistry Knowledge Graph based on the specified requirements. This provides structured, topic-relevant context for question generation.

3) Prompt Construction Agent: The Prompt Construction Agent combines the structured requirements with the retrieved Knowledge Graph information to construct a contextual prompt for the LLM. The prompt includes the selected topic, relevant concepts, difficulty constraints, and question-generation instructions.

4) Question Generation Agent: The Question Generation Agent uses the LLM and the constructed prompt to generate Chemistry MCQs with answer options, correct answers, and explanations. It also supports image-based question generation by creating visual representations of relevant chemical structures and diagrams.

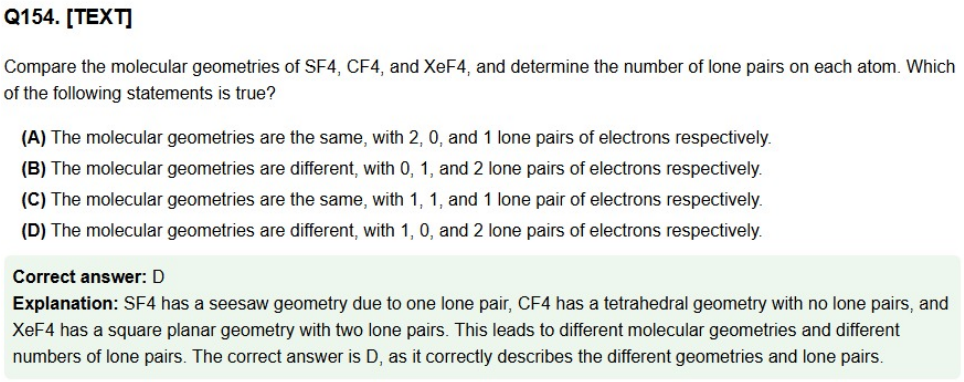
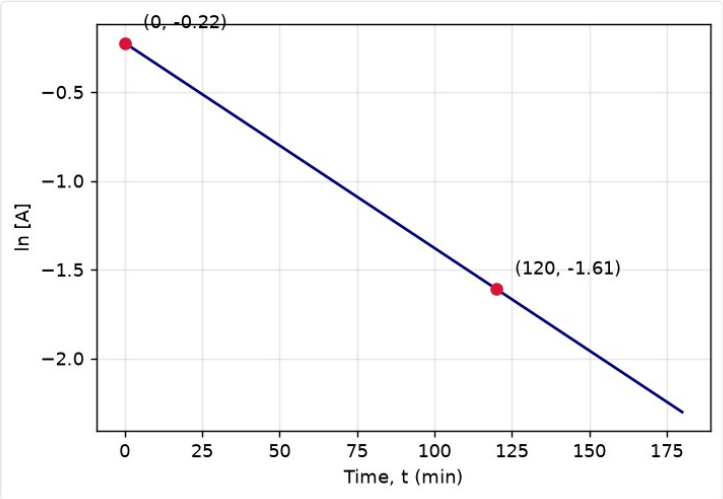


Figure 5: *Text-based Chemistry question.*

Q39. [IMAGE]

The graph shows ln[A] versus time for a first-order reaction A → products. The half-life of the reaction, estimated from the two marked points, is closest to



(A) 30 min
(B) 60 min
(C) 86.6 min
(D) 120 min

Correct answer: B
Explanation: Slope = $(y_2 - y_1)/(t_2 - t_1)$ = (-1.61 − (-0.22))/120 = −0.0116 min^{-1}, so k ≈ 0.0116 min^{-1}. $t_{½}$ = 0.693/k ≈ 59.8 min ≈ 60 min.

Figure 6: *Image-based Chemistry question involving a reaction graph.*

Q198. [TEXT]

For the reaction H2(g) + I2(g) ⇌ 2HI(g), the equilibrium constant Kc decreases from 66.9 to 50.0 when the temperature increases from 350°C to 448°C. What does this indicate about the enthalpy change of the reaction?

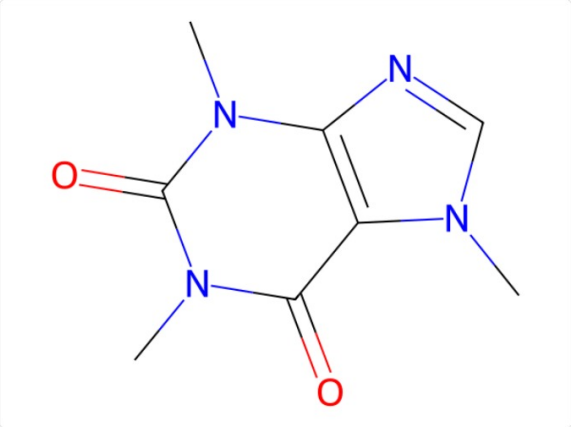
(A) The sign of ΔH cannot be determined
(B) ΔH is positive
(C) ΔH is negative
(D) ΔH is zero

Correct answer: C
Explanation: According to Le Chatelier's principle, if increasing the temperature decreases the equilibrium constant (Kc), the reaction must be exothermic (ΔH < 0). This is because the system shifts to counteract the increase in temperature by favoring the reverse (endothermic) reaction, which reduces the value of Kc. Hence, the enthalpy change is negative.

Figure 7: *Text-based Chemistry question on chemical equilibrium.*

Q54. [IMAGE]

How many σ bonds (counting every bond to hydrogen as well) are present in the structure shown?



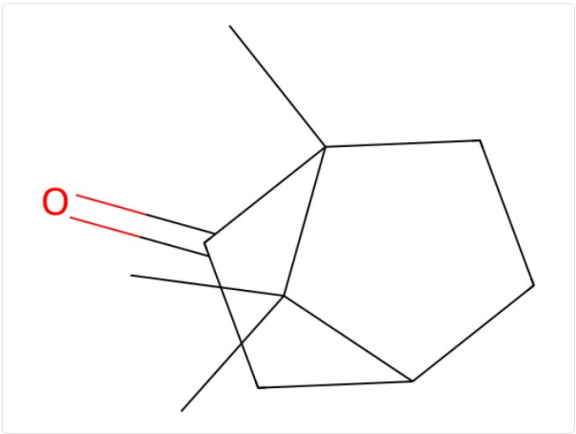
(A) 15
(B) 24
(C) 25
(D) 26

Correct answer: C
Explanation: The structure shown is caffeine (C8H10N4O2). Every single bond is one σ bond, a double bond has one σ and one π, a triple bond has one σ and two π. Counting all bonds including those to hydrogen gives 25 σ bonds (15 without bonds to H).

Figure 8: *Image-based Chemistry question involving a chemical structure.*

Q49. [IMAGE]

The total number of rings present in the structure shown is



(A) 0
(B) 1
(C) 2
(D) 3

Correct answer: C
Explanation: The structure shown is camphor (C10H16O). Counting the smallest set of smallest rings gives 2 rings.

Figure 9: *Image-based Chemistry question involving a ring structure.*

5) Similarity Checking Agent: The Similarity Checking Agent uses a Swin Transformer-based approach to compare image features and identify duplicate or highly similar images associated with Chemistry questions. This helps reduce image-level repetition and improve the diversity of the generated question set.

This design mirrors human exam-setting practice, extended into a fine-grained agent pipeline: identify

Table 1: Multi-Agent System — Responsibilities, Inputs, and Outputs

Agent	Responsibility	Input	Output
1. Requirement Analysis Agent	Identifies the topic, difficulty, question type, and question count.	User request	Structured requirements
2. Knowledge Graph Retrieval Agent	Retrieves relevant questions and concepts from the Chemistry Knowledge Graph.	Topic and requirements	Relevant questions and concepts
3. Prompt Construction Agent	Builds a structured LLM prompt using retrieved knowledge and requirements.	Requirements and KG data	Generation prompt
4. Question Generation Agent	Generates Chemistry MCQs using an LLM and creates chemical structure images using RDKit and Matplotlib.	Prompt and KG context	Questions, options, answers, explanations, and images
5. Similarity Checking Agent	Checks generated questions against existing questions to identify duplicates and highly similar content.	Generated questions and dataset/KG	duplicate status

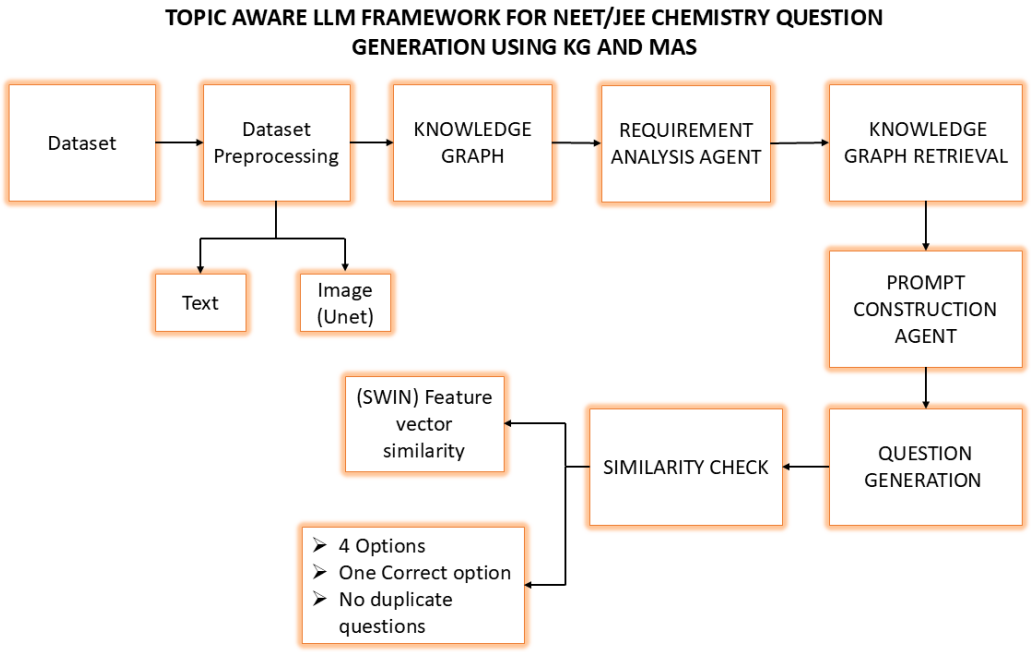


Figure 10: *Proposed workflow of the knowledge graph-based multi-agent framework for topic-aware NEET/JEE Chemistry question generation, similarity checking.*

the syllabus topic, pull supporting evidence from the Knowledge Graph, extract the governing concepts, calibrate difficulty, assemble a structured prompt, generate the question, check it against existing items for duplication, validate its quality and format, and route any failures back through the Feedback Agent for regeneration. The agents do not replace the LLM; they constrain, sequence, and verify it using structured

G. LLM-Based Question Generation

The Prompt Construction Agent builds a structured prompt containing:

1. the target chapter and topic,
2. relevant questions and concepts retrieved from the Knowledge Graph,
3. the desired difficulty level, and
4. formatting requirements for NEET/JEE MCQs.

The Question Generation Agent uses the Qwen3-8B model with 4-bit quantization to process the constructed prompt and retrieved knowledge and generate Chemistry MCQs, including the question stem, four options, the correct answer, and an explanation. For image-based questions, the agent also generates relevant visual representations of chemical structures and diagrams. The Similarity Checking Agent uses a Swin Transformer-based approach to identify duplicate or highly similar images associated with the generated questions. This helps reduce repetition and improve the diversity of the generated question set. The overall framework combines Knowledge Graph-based retrieval and Qwen3-8B-based question generation to produce topic-aware questions aligned with the specified requirements.

H. Swin Transformer

Swin Transformer Tiny is used to extract hierarchical visual features from the preprocessed Chemistry images. Each image is resized to 224×224 pixels, converted into a three-channel grayscale image, and normalized using ImageNet mean and standard deviation values. The pretrained Swin Transformer processes each image through four hierarchical stages using shifted-window self-attention.

The four stages generate feature maps with dimensions $56 \times 56 \times 96$, $28 \times 28 \times 192$, $14 \times 14 \times 384$, and $7 \times 7 \times 768$, respectively. As the network progresses, the spatial resolution decreases while the number of feature channels increases, enabling the extraction of visual information at multiple scales.

The final feature map from Stage 4 is processed using **global average pooling**, which converts the $7 \times 7 \times 768$ representation into a 768-dimensional vector. This vector is passed through a multilayer perceptron (MLP) head consisting of three fully connected layers with dimensions $768 \rightarrow 512 \rightarrow 256 \rightarrow 128$. ReLU activation is applied after the first two layers, and the final layer produces a **128-dimensional feature vector** for each image.

These feature vectors provide numerical representations of the Chemistry images and are used for pairwise similarity comparison and potential duplicate detection. The Swin Transformer is frozen during feature extraction. The MLP head in the current implementation is randomly initialized and untrained; therefore, its output represents a 128-dimensional projection rather than a task-trained embedding.

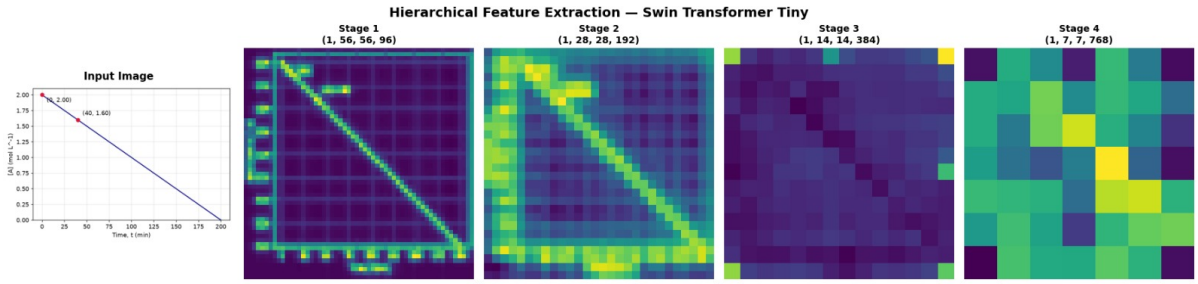


Figure 11: *Hierarchical Feature Extraction Using Swin Transformer Tiny.*

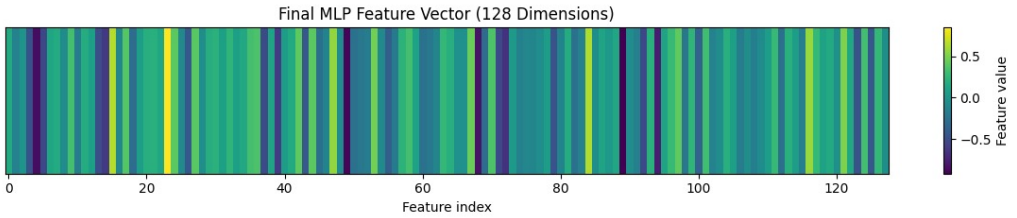


Figure 12: *128-Dimensional Feature Vector Generated by the MLP Head.*

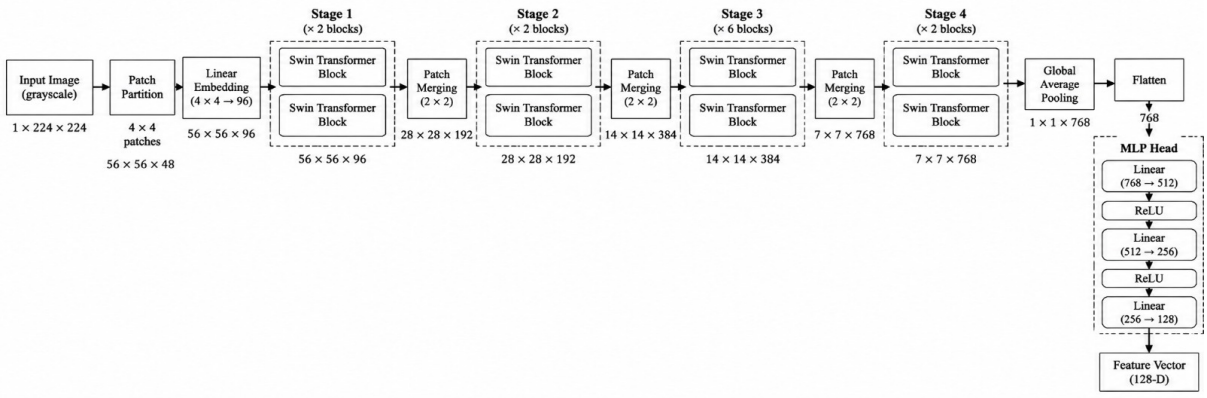


Figure 13: *Swin Transformer Tiny Architecture and MLP-Based Feature Projection.*

I. Sinkhorn Optimal Transport for Image Similarity

The feature vectors extracted from Chemistry images are compared pairwise using Sinkhorn Optimal Transport. The squared Euclidean distance between two 128-dimensional feature vectors is used as the cost matrix, with an entropy regularisation parameter of 0.05. The transport cost is converted into a similarity score using

$$S = \frac{1}{1 + D},$$

where D represents the transport cost. All image pairs are ranked according to their similarity scores, and highly ranked pairs are flagged as potential duplicates for manual review.

From the pairwise similarity analysis, approximately **19–20 image pairs were identified as similar** based on the selected similarity threshold. These pairs can be reviewed to determine whether they contain duplicate or highly similar Chemistry questions.

Note: Since each image is represented by a single feature vector, the current implementation mathematically reduces to squared Euclidean distance rather than performing meaningful patch-level optimal transport. Therefore, the identified pairs should be treated as potential similarities requiring further validation.

Step 1: Feature Vector Normalization

Each image is represented by a 128-dimensional feature vector X_i, which is normalized to unit length before pairwise comparison:

$$\hat{X}_i = \frac{X_i}{\|X_i\|_2} \tag{1}$$

Step 2: Pairwise Cost Calculation

The squared Euclidean distance between two normalized feature vectors is calculated as:

$$C_{ij} = \sum_{k=1}^{128} \left(\hat{X}_{ik} - \hat{X}_{jk} \right)^2 \tag{2}$$

A smaller cost indicates greater similarity between the feature vectors.

Step 3: Sinkhorn Optimal Transport

Each image is represented by a single feature vector with unit mass. Therefore, the source and target distributions are:

$$a = [1], \qquad b = [1], \qquad P = [1] \tag{3}$$

The resulting transport cost is:

$$D_{\mathrm{OT}} = P \times C_{ij} = C_{ij} \tag{4}$$

Step 4: Similarity Score Calculation

The transport cost is converted into a similarity score using:

$$S_{ij} = \frac{1}{1 + D_{\mathrm{OT}}} \tag{5}$$

A smaller transport cost produces a higher similarity score, while a larger cost produces a lower score. For example, if $D_{\mathrm{OT}} = 0.25$, then:

$$S_{ij} = \frac{1}{1 + 0.25} = 0.8 \tag{6}$$

Thus, the similarity score is 0.8.

Implementation Limitation: Since each image is represented by a single feature vector, the current implementation effectively reduces to squared Euclidean distance. Meaningful patch-level optimal transport would require multiple local feature vectors per image.

Conclusion

The proposed system successfully generated 200 chemistry questions using Qwen as the Large Language Model (LLM). U-Net was employed for image preprocessing and denoising to improve the quality of chemistry question images. The Swin Transformer was used to extract visual feature vectors, which were then used for similarity analysis using Sinkhorn Optimal Transport. When a generated question exceeded the defined similarity threshold with an existing question, it was selected for regeneration to minimise duplication. This integrated approach combines automated question generation, image preprocessing, visual feature extraction, and similarity-based validation. The system aims to improve the quality and diversity of generated chemistry questions while reducing repetitive content.

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